

ML Problem Solving

Vector arithmetic and linear classifier inference

1. A linear model $\hat{y} = a_1x_1 + a_2x_2 + b$ has $a = [-1, 1]$ and $b = 2$. For the following x , determine the score $\hat{y}$ and the *associated prediction*. Recall that our linear classifier was trained on targets $t \in \{0, 1\}$.

a) $[[0], [0]]$

b) $[[-1], [1]]$

c) $[[1], [-1]]$

d) $[[1.5], [0]]$

e) $[[0], [-1.5]]$

2. Consider that we want to represent $a \cdot x + b$ as a single vector multiplication $a \cdot x$, without explicitly adding the bias b , where a and x are vectors and b is a scalar. What would be the values for a and x for each input?

a) $a = [\text{ ______ } , \text{ ______ } , \text{ ______ }] , x = [\text{ ______ } , \text{ ______ } , \text{ ______ }]^T$

b) $a = [\text{ ______ } , \text{ ______ } , \text{ ______ }] , x = [\text{ ______ } , \text{ ______ } , \text{ ______ }]^T$

c) $a = [\text{ ______ } , \text{ ______ } , \text{ ______ }] , x = [\text{ ______ } , \text{ ______ } , \text{ ______ }]^T$

d) $a = [\text{ ______ } , \text{ ______ } , \text{ ______ }] , x = [\text{ ______ } , \text{ ______ } , \text{ ______ }]^T$

e) $a = [\text{ ______ } , \text{ ______ } , \text{ ______ }] , x = [\text{ ______ } , \text{ ______ } , \text{ ______ }]^T$

3. Now assume we want to calculate all 5 scores in a single matrix multiplication. Write down the matrices and vectors for this data and model. What do we need to change compared to our linear model $a \cdot x$?

4. Draw a coordinate system with datapoints a to e. Draw the decision boundary of linear model $\hat{y}$.

5. Now let's assume we have a new $b = 0$ with a unchanged. Draw the decision boundary of the new model.